QGIS Plugin Overview: Land-Use Continuity Detection

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Link: github.com/Rintozxr/landuse-continuity-qgis-plugin

Demo Video: Detecting Land Use Continuity in QGIS – Plugin Demo with Amsterdam Data (youtube.com)

Overview

The Landuse Continuity Plugin is a lightweight, open-source QGIS plugin designed to assist urban researchers, planners, and historians in identifying areas of spatial form persistence across geo-referenced historical land-use maps. It helps answer critical questions in urban history and public health: Which urban parcels have remained stable over time? Where do changes in spatial configuration correlate with shifts in social or infrastructural function?

This tool was developed as part of a methodological exploration for a PhD project proposal on Mapping Historical Blue Infrastructures for Urban Health. It is particularly suited for diachronic analysis of archival vector maps, such as those depicting canal systems, disease geographies, or planning typologies.

Key Features

Automatic spatial matching between multiple polygon layers (e.g., land-use maps from different years), identifying parcels that persist across time periods.

Flexible threshold setting to determine the minimum geometric overlap required to count as "continuous".

Interactive UI built with PyQt5 and Qt Designer, allowing users to:

- Select relevant input layers
- Adjust comparison thresholds
- Monitor analysis progress via progress bar

Attribute tagging: The base layer is automatically updated with new boolean or integer fields indicating continuity classification.

Null-safe, attribute-agnostic comparison: Only spatial geometry is used, making it suitable for historical datasets with inconsistent metadata.

Example Use Case

In the demonstration project, the plugin was used to compare 2006 and 2018 Urban Atlas land-use layers for northwestern Amsterdam.

Step 1: Prepare the Input Data

We use publicly available land-use maps from Copernicus Urban Atlas, containing polygonal representations of Amsterdam's land-use parcels in 2006 and 2018. Both layers are loaded into QGIS as shapefiles.

- Layer 1: Amsterdam2006sample.gpkg
- Layer 2: Amsterdam2018sample.gpkg

Step 2: Launch the Plugin

Open the Landuse Continuity plugin via the QGIS Plugins menu. A simple user interface appears (Figure 1), allowing users to:

- Select two or more polygon layers through Layer List
- Set the continuity threshold (e.g., 0.8 = 80% area overlap)
- Monitor progress via the real-time progress bar

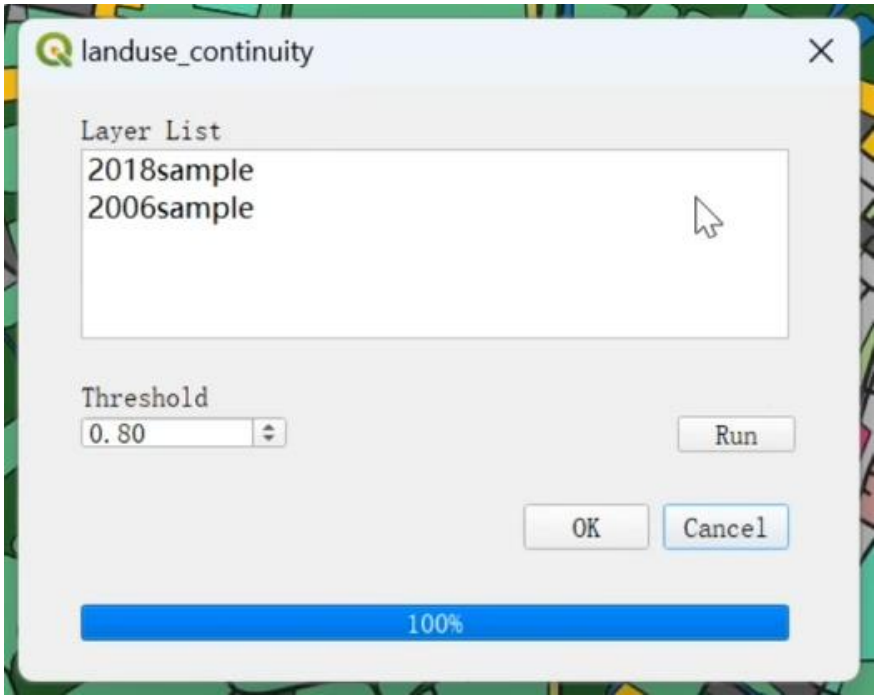

Figure 1. User Interface of the Landuse Continuity Plugin

Step 3: Run the Analysis

Click the "Run Analysis" button. The plugin performs the following actions internally:

- Iteratively compares all features in the base (earlier) layer with the newer layer
- Calculates pairwise area intersection ratios
- Flags polygons that remain geometrically similar as "True" in a new field: is continuous (Figure 2.)

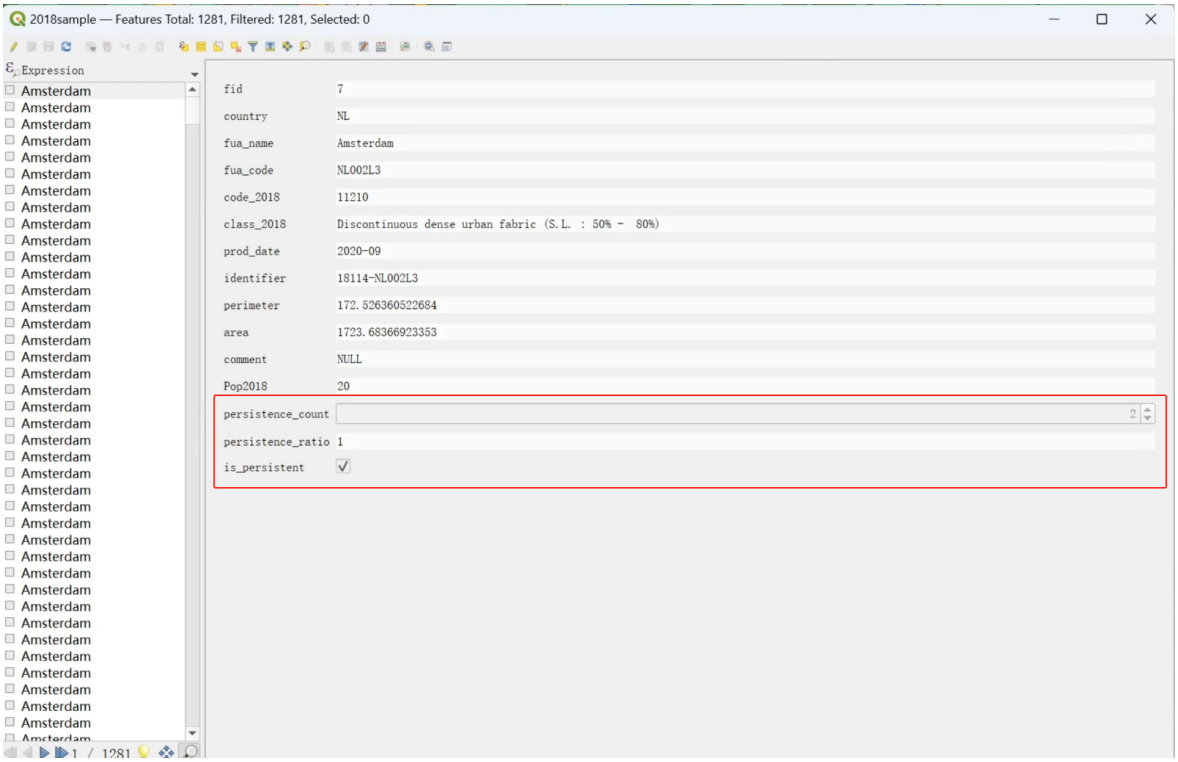

Figure 2. Newly Added Continuity Fields in the Attribute Table

Step 4: Visualize the Result

The plugin modifies the latest layer (e.g., 2018) by appending a is continuous field to each polygon. Users can then style the layer using categorized symbology to highlight:

- True – Parcels with strong geometric continuity

- False – Parcels that have changed shape or disappeared

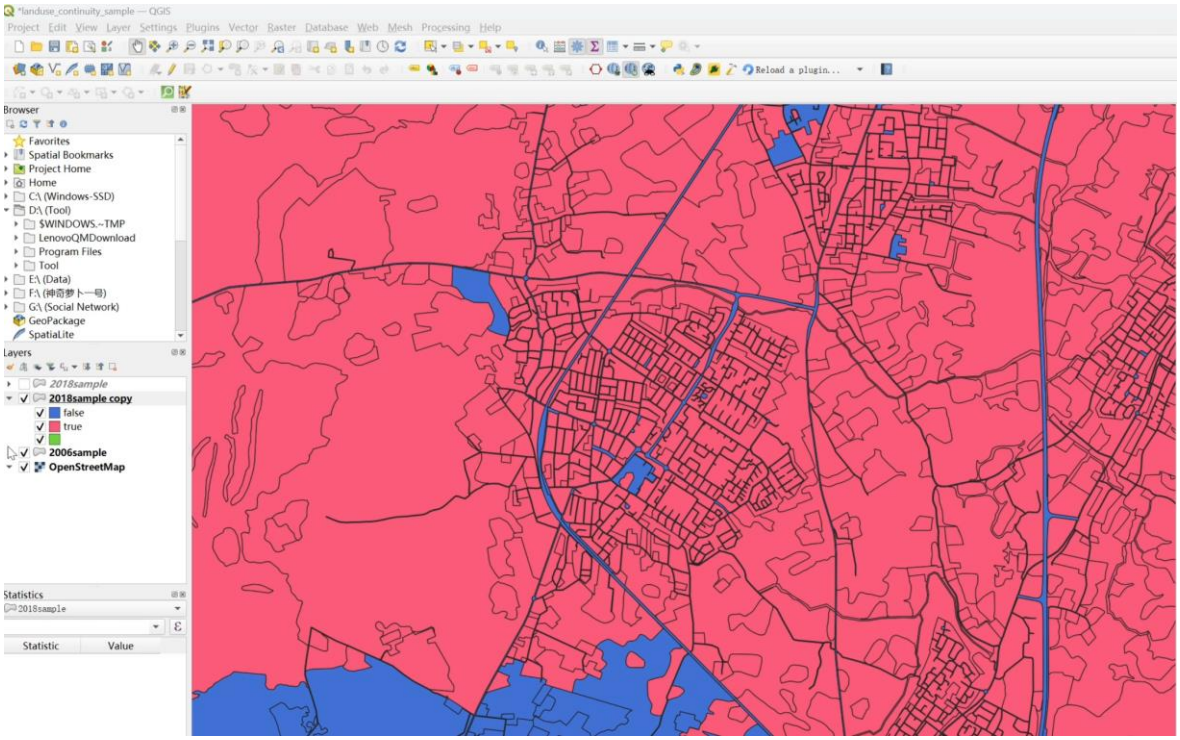

Figure 3. Visual Differentiation of Stable (True) and Changed (False) Parcels Based on Continuity Calculation (Red polygons = stable parcels; Blue polygons = modified or missing parcels)

Why This Matters

Mapping land-use continuity over time offers unique insight into how urban form durability underpins broader shifts in infrastructure function, public space utilization, and urban health outcomes. By providing a simple yet powerful way to identify persistent spatial structures, this plugin supports interdisciplinary research bridging urban history, planning, and digital humanities.

Development Context

This plugin was developed independently as part of a broader methodological exploration during PhD research preparation. It originated from an effort to investigate diachronic spatial transformation through lightweight, reproducible digital tools. While initially designed to assess historical land-use continuity along urban waterfronts, its underlying logic, based on geometric persistence, offers broader applicability in studying ruptures, voids, and infrastructural reconfigurations across time. The tool reflects my commitment to integrating spatial computation with critical urban inquiry and has been released as an open-source resource to support interdisciplinary research at the intersection of urban history, planning, and digital humanities.